

User Experience Impact of Various User Interfaces on VR Laparoscopy Tasks: A Preliminary Study

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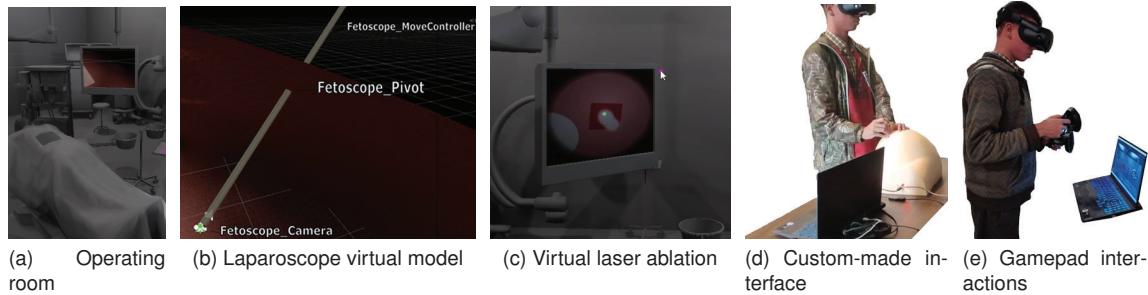


Figure 1: Virtual laparoscopy setup showcasing the virtual environment and two user input devices.

ABSTRACT

Recent advances in immersive technologies have facilitated the development of highly realistic and cost-effective computer-based simulations (CBS) and this has greatly influenced health professions training. Unlike high-end simulators that require specialized devices, keyboards, mice, gamepads, virtual reality (VR) controllers, and 3D-printed custom interfaces facilitate interactions with CBSs. However, given the lack of proper task representation, many questions remain regarding their effects on the user experience. This paper presents usability, cognitive load and presence preliminary study comparing a virtual laparoscope manipulation using a keyboard and mouse, gamepad, VR controllers and a 3D-printed laparoscopic controller.

Index Terms: Virtual reality, laparoscopy, user input devices.

1 INTRODUCTION

Laparoscopy is a minimally invasive surgery (MIS) that allows a surgeon to access the interior of the abdomen, eliminating the need for large incisions in the skin, leading to a faster recovery time, fewer postoperative complications, and fewer life-threatening risks [1]. Once inserted, the laparoscope is manipulated to locate landmarks of interest, with laser activation pressing a foot pedal. Laparoscopy training has seen the availability of high-end simulators that employ robotics for haptics and force feedback in conjunction with computer graphics to properly represent visual and tactile tasks [7]. However, high-end simulators are costly, require technological

updates and maintenance, and their access depends on availability and scheduling [2]. Such drawbacks have sparked the development of low-cost simulators using consumer-level VR, which, despite the lack of proper medical instrument representation, are being used for medical training and education, with many questions remaining regarding their effects on the user experience, in particular, pertaining to more accurate user input devices, such as those being developed using 3D printing. For example, a custom-made 3D-printed laparoscope for computer-based simulation training [4], performing procedures using VR controllers and hand tracking [5].

This paper presents a preliminary study that examined how keyboard and mouse, gamepad, VR controllers, and a custom-made 3D-printed laparoscope impact usability, cognitive load, presence, and heart rate when operating a virtual laparoscope. Our motivation stems from the high adoption of VR as a training tool for medical procedures and the use of user input devices lacking adequate instrument representation.

2 MATERIALS AND METHODS

A within-subjects study approved by Ontario Tech University's Research Ethics Board #17127 required participants to complete aim and laser tasks manipulating a virtual laparoscope using keyboard and mouse, a gamepad, a VR controller, and a custom-made 3D-printed laparoscope.

2.1 Study Procedures

Participants completed the following for each condition: i) familiarizing themselves with the keyboard/mouse, VR controller, gamepad, custom-made laparoscope, and pedal for manipulating the virtual laparoscope and triggering its laser based on the randomly assigned condition, ii) manipulating the laparoscope to locate and laser anatomical landmarks depicted by squares while avoiding circles, and iii) completing the System Usability Scale (SUS) [3], the raw NASA Task Load Index (RTLX)[6], and the Virtual Reality Presence Questionnaire (VRPQ) [8], followed by an open question to provide feedback.

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2.2 Apparatus

The laparoscopic simulator was developed using the Unity game engine (2021.1.5f1) for the HP Omnicept VR headset, which supports heart rate (HR) measurements without requiring external devices. The scenario presented an operating room with a monitor that shows the view of the laparoscope camera (Fig. 1), and the virtual laparoscope already in place. The laparoscope has two rotational degrees of freedom (DOF) to change the camera view left/right (yaw) and up/down (pitch), and one translational DOF to move the laparoscope in/out of the abdomen. An abstract representation of the vessels and moving obstacles using squares and ovals, respectively, was chosen for novices. A pedal was integrated to provide realistic laser activation, and a shader was used to depict the ablation caused by the laser photocoagulation when triggered. The scenario requires locating and cauterizing targets, which turn green when completed. The virtual laparoscope was operated using a wired mouse and QWERTY keyboard (W and S keys for translation) and a Logitech MX vertical mouse for camera rotation), a wired Xbox360 gamepad and the HP Omnicept VR controllers used similar inputs (i.e., the left joystick for translation and the right joystick for camera rotation), and the 3D-printed laparoscope required physical pitch, yaw, and penetration manipulation.

3 PRELIMINARY RESULTS

Thirteen participants (male = 7, female = 6) with backgrounds in Health Sciences (6), Computer Science, and Information Technology (7) volunteered. All participants reported having some VR experience and being active console and computer video game players. None of the participants had previous laparoscopic experience. The results of VR laparoscopy tasks are summarized in Fig. 2. Regarding usability, scores above 68 are considered acceptable and scores above 80 are excellent. In terms of usability, the keyboard and mouse (SUS = 81.25) had the highest score, followed by the VR controllers (SUS = 78.75), the gamepad (SUS = 77.5), and lastly, the custom-made 3D-printed laparoscope (SUS = 74.64). Regarding cognitive load, the raw NASA TLX (RTLX) scores between 0 and 29 indicate a low to medium workload, with the keyboard and mouse having the lowest score (RTLX = 18.30), followed by a tie between the gamepad and VR controllers (RTLX = 21.60), with the custom-made 3D-printed laparoscope last (RTLX = 23.30). Presence scores (VRPQ), the higher, the better, indicate that the custom-made 3D-printed laparoscope had the highest score (VRPQ = 89.80), followed by the keyboard and mouse (VRPQ = 84.50), the VR controller (VRPQ = 81.50), and finally, the gamepad (VRPQ = 77.50). During task execution, heart rate monitoring indicated that the custom-made 3D-printed laparoscope had the highest average beats per minute (HR = 91.09 bpm), followed by the VR controller (HR = 71.43), the keyboard and mouse (HR = 62.11), and the gamepad (HR = 34.59). It should be noted that the lowest HR value was below the average for a healthy person (60-100 bpm) caused by data loss, possibly due to connectivity issues.

4 DISCUSSION

This paper presented a preliminary user study that focused on evaluating the impact of various input devices when performing virtual laparoscopic tasks on the user experience. The custom-made 3D-printed laparoscope input device resulted in lower usability, higher cognitive load, and higher presence scores compared to all other input devices. We believe that these results reflect the unfamiliarity with manipulating the laparoscope input device by the participants; in contrast, the other, more common devices had higher usability and lower cognitive. Interestingly, despite the lower usability and high cognitive load, the custom-made 3D-printed device increased the sense of presence. Based on participant feedback such as “the simulation was engaging and immersive” and “The clinic made the task feel real.” We believe that the realism of the custom-made

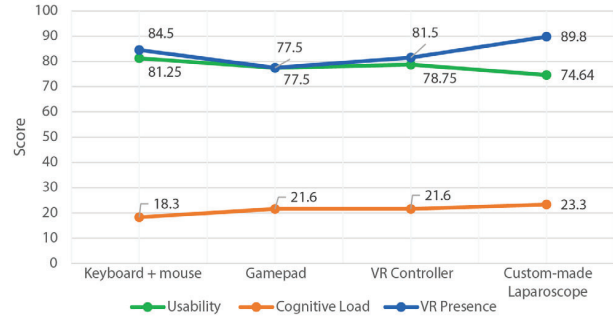


Figure 2: Usability, cognitive load and presence, results.

3D-printed laparoscope added to the sense of presence. Regarding heart rate, the custom-made 3D-printed laparoscope had a higher HR than the other input devices. This could be attributed to the difficulties of manipulating an unknown device for the first time when trying to complete the task. However, to identify the factors that contribute to elevated HR, further investigation is needed. In conclusion, this preliminary study shows that proper representation of the user input device can impact presence at the expense of usability and cognitive load.

Future work will focus on a larger user study with trainees and experts, furthering our understanding of the effects that input devices have on knowledge retention and motor skills development.

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